

UNIT - IV

METHODS OF TEACHING

Methods of teaching

A method is "a scientific way of presenting the subject keeping in mind the psychological and physical requirements of children."

A History Teacher has a variety of methods and techniques available for the use in teaching History. The selection of suitable method depends upon the objectives of the lesson, nature of the content and needs of the learner. Some methods are more appropriate for teaching students as a group whereas some methods are specially designed for individual instruction. A method best for one teacher and applicable for a class, may totally be a failure for another teacher to teach the same other class. So it should be left for the teacher from his wide information, which of the methods to use and when. Here we are going to discuss in detail the various methods available to teach History.

Teacher-centred methods

In these methods the teacher's role is to give knowledge and full control of the class room.

4.1 Lecture Method

It is the oldest teaching method. It is the most dominating method and is liked by the majority of

Teachers. In this method teacher present the subject matter and the students listens. Students are not participating in this method. This method is teacher centred and information centred and in this method teacher works as a sole resource in classroom induction. Due to lack of participation, students get bored and some of them may go to sleep. Here teacher is presenting ready made knowledge and due this spoon feeding. The students loose interest and their power of reasoning and observation get no stimulus.

In this method teacher goes ahead with the subject matter at his own speed. The teacher may use blackboard at times and may also dictate notes. This teacher centred method in its extreme form does not expect any question or response from the students. It helps only intelligent students who can understand and remember the facts told by the teacher, which is just a blind exercise.

In this method, teacher prepares his lesson plan at home and delivers it out in the class. This method does not cater for releasing the teaching History and is not in accordance with the principles of teaching.

Guidelines for preparing a Lecture

To make lecture method effective, the teacher has to follow certain guidelines while preparing the lecture.

- Lecture should be carefully and systematically planned as lesson plan

- Teaching objectives should be kept in mind while preparing the lecture.
- The lecture should exhibit the through knowledge of the subject matter, its organisation, development, interpretation and application.
- Lecture should be related with day to day life experience of the students.
- Student participation should be encouraged by asking questions, clarifying doubts and reviewing the key points at frequent intervals.

Merits of Lecture method

1. It is possible of handle a large number of students at a time and no laboratory, equipment, aids are required.
2. Using this method the knowledge can be imparted to the students quickly and the prescribed syllabus can be covered in a short time.
3. Teachers feel secure and satisfied.
4. Using this method it is quite easy to impart factual information, historical anecdotes and life history of eminent personalities.
5. Teacher can develop his own style of teaching.
6. In this method teacher can easily the logical sequence of the subject by planning his lecture.
7. This method gives the students as well as the teacher a sense of satisfaction and achievement.

8. Lecture method trains students to listen long duration.
9. It delivers a lot of information in a short amount of time.

Demerits of Lecture method

1. In this method students participation is negligible and students become passive recipients of information.
2. We are not sure if the students are concentrating and understanding the subject matter.
3. This method leads weak students develop a hatred for learning.
4. It does not take into account the previous knowledge of the students.
5. It does not cater to the individual needs and differences of the students.
6. It is the lowest retention value of all teaching techniques.
7. It is an undemocratic and authoritarian method.
8. Inability to understand one essential point may make the rest of the lecture unintelligible.
9. This method does not help in developing problem solving skills.
10. This method can be used only in higher secondary level or college level.

4.2. Problem solving method

Problem solving method is one which involves the process of reflective thinking to solve a different problem in a purposeful manner and in arriving a satisfactory solution.

The problem solving method is like a research method. It involves scientific thinking of a process of learning.

Steps in problem solving

1. Identifying problem solving.
2. Collecting the data.
3. Formulating tentative hypothesis.
4. Testing the hypothesis.
5. Verification of the result.

1. Identifying the problem

The students should be able to identify and clearly define the problem and recognize the type of the problem. The problem should be interesting, challenging and motivating for the students to solve.

2. Collecting data

What is given in the problem ? What is asked in the problem? Whether the given data is sufficient? Analyse the problem carefully and collect information to solve the problem.

3. Formulating tentative hypothesis

Analyse the data and find the interrelationships among the given facts. The student formulates tentative solution of the problem.

4. Testing the hypothesis

Appropriate methods should be selected to test the validity of the tentative hypothesis as a solution to the problem. If it is not proved to be the solution, the students are asked to formulate alternative hypothesis and proceed.

5. Verification of the result

In Physical Science, we do not accept a solution without proper verification. The students are required to verify the solutions and make generalizations and apply them in a new life situations.

Teacher's Role in Problem solving method.

The teacher plays a important role in problem-solving method.

- give an atmosphere of freedom, defining and stating the problem.
- assist the students in accepting, defining and stating the problem.
- help the students in breaking up the problem into simple units.
- guide the students in locating relevant source materials.
- help the students to develop spirit of enquiry and discovery.

Merits of Problem-solving method.

1. Problem solving provides a real life experience to the students.

2. It develops in students good habits of planning, thinking, reasoning and independent work.
3. This method serves individual difference.
4. It develops qualities of self dependence and self confidence.
5. The students get valuable social experiences such as patience, co-operation, helping others etc.

Demerits

1. Not all the students are problem solvers.
2. It is time consuming and consequently it is not possible to cover the syllabus on time.
3. Reference and resource materials may be difficult to get.
4. The success of this method depends upon Physical Science teachers who are well versed in critical thinking and reflective thinking.
5. This method does not suit the students of lower classes.

4.3. Biographical method

Biographical method is the intensive collection and analysis of the events of an entire life or part of life. This can be reinforced by an in-depth unstructured interview in general.

Biographies in general play an important role in examining a person's success and knowing how he or

she achieved his or her goals. Biographies also reveal where and why they failed. Biographies illustrate the commonalities of all leaders. Autobiographies are a way to learn our history and provide a compelling story. An important part of knowing our history is developing a better understanding of our political leaders. Our leaders are somewhat flawed. Considering them to be gods would actually cause a backlash. Our leaders have struggled to achieve extraordinary things. This is very encouraging among us. We have a lot to learn from finding great leaders and their experiences. Autobiographies play an important role in examining the successes of past leaders and how they achieved their goals. All the failures and shortcomings they have experienced in life will help them to understand.

Therefore, good autobiographies should change our understanding of the past and give us a full understanding of the past. A good autobiography should provide plenty of color about everyday life. Only then will it make people interested and worth reading.

Types of autobiographies

Autobiography is a special form of history. It is a collection of events based on the example of a person's life. There are four basic types of autobiography. They are

1. Biography of historical fiction
2. Academic Biography
3. Imaginary education
4. Biography of the visionary to solve

1. Biography of historical fiction

Imaginary style is often used in the biographies of living celebrities, athletes, and politicians. Such stories often inspire stories for film or television. Are based on some basic facts about the individual and are then created for better entertainment value. Here there is little concern for the direct accuracy or truth about the life lessons of the individual.

2. Academic Biography

The second type of historical story based on individual human life is academic biography. Academic autobiographies rely heavily on documented facts and significant achievements in a person's life. Historians depict relevant facts around a person's achievements. The lives of leaders in political and social change are generally summarized chronologically in an academic biography. It is full of notes that refer to detailed notes. These autobiographies have a limited audience.

3. Fictional academic biographies

The third type of autobiography is fictional academic biography. This is a fictional academic biography. This time it tries to take the best elements of the autobiography and combine it with the entertainment. It provides a balanced view of how one might have lived by combining the author's unique views of life with the facts and lessons of the individual.

4. Biography of the visionary to solve

A fourth type of prophetic biography is a review of a person's life story. The prophetic biography begins with an educational approach that considers all known facts. Once the details are listed a spiritual goal or liberation of the people is created and facts are added to achieve the best entertainment value. The prophetic biography differs from the fictional entertainment autobiography. Here the goal is to communicate practical life lessons for the overall improvement of the reader. The autobiography of Mohandas M. Gandhi "" The Story of My Trials True "" is an example of this. Many of the world's leading literary works are autobiographies, including Julius Caesar's Commentaries, Benjamin Franklin's Autobiography, and St. Augustine's Confessions of St. Augustine.

Advantages of the biographical method

1. Its influence motivates and motivates us as we read the real life story of someone who is successful or notable. The student uses it to solve problems in his or her own life by realizing how they have overcome the trials of others' lives.
2. Learning from the real experience of others is a great tool to help students grow into members who contribute to a successful community.
3. Biographical method is used to maintain a reasonable view and to recognize the point of view of others.

4. The concept of having to work diligently to achieve goals benefits students.
5. We learn to be loving, helpful and caring towards others.

Disadvantages of the biographical system

1. The historian will insert his own thoughts into the autobiography.
2. The historian also has the opportunity to edit real life events.

4.4 Story-telling Method

Story-telling can teach ethics, values and cultural norms and differences. Teaching through this method in social contexts will yield good results. Stories are used as excellent tools to carry knowledge in social contexts.

Story-telling is the process of explaining the social and cultural function of sharing stories. Sometimes it is shared as a way for each culture to explain its own stories. It is used for aspects such as education, cultural preservation, moral values or entertainment. The key elements of storytelling are the plot, story perspective and characters. Storytelling often refers to the oral method of storytelling.

Learning through story-telling

Story-telling is a way to share and explain experiences, says Beta L. Berger said. Human life is rooted in the story. Stories related to culture, linguistics and age are global. We can leave the concept of age

and tell the story to all ages. Human knowledge is based on stories and the human brain has the knowledge needed to understand, remember and tell stories. Stories reflect human thinking. Humans think in story structure and often remember facts in story form. Story-telling supports analytical thinking.

Story-telling is a useful educational tool

Stories are useful educational tools. Story-telling can be seen as the foundation for learning and teaching. It allows the individual to be actively involved in the story and to listen, listen and participate, with minimal guidance. Story-telling can inspire innovative problem solving and develop an understanding of future ambitions. Story-telling is used as a tool to teach children through listening practice. Stories are used to respect everyone and work together.

Story-telling helps students to understand other cultures and to develop a positive attitude towards people of different nationalities, races and religions.

Four types of story-telling

There are many approaches to story-telling. Here we look at four important types of story-telling. They are

1. Personification
2. Third party story-telling
3. Embedded journalism
4. Metaphors

1. Personification

If the narrator acts in a way that is most intriguing he and his audience can share experiences together. We identify the general character of ideological content by sharing an event, an idea, or a reflection.

2. Third party story-telling

Third-party storytelling is about telling people how their lives have become better or richer by telling an idea.

3. Wood packaged journalism

This approach seeks to provide a journalistic view of an organization or related external events and happenings.

4. Metaphors

Embedding metaphors in content makes it easier to imagine difficult ideas. Usually through the use of metaphors, the ability to identify the ideas behind a story is realized.

The role of the teacher in story-telling

- Design the stories according to the subject matter.
- Legends, true stories, myths, and stories from magazines can be told if the subject matter is appropriate.
- The teacher should tell the story in simple words packed with meaning that students can understand. Stories should be intriguing.

- The lesson becomes engaging as the teacher expresses laughter, joy, affection, compassion, friendship, heroism with voice fluctuations of emotions with the characters.

Advantages of the story-telling method

1. Storytelling involves many media.
2. Expands students' knowledge and cultural beliefs and activities.
3. Establishes good relationship between teacher and students.

Disadvantages of story-telling

1. If the narrator does not have the required skills, the chance of learning decreases.
2. The story should be able to be completed during the respective course without being too long.

4.5 Discussion Method

Discussion is one of the most valuable techniques in the teaching of history. It aims at finding the solution of a problem through the establishment of agreement or consensus. Discussion is a sharing and weighing of the sides; which are as many as there are conflicting interests and opinions. It is a process of collective decision making. It seeks agreement, but it is not reached, it has the value of clarifying the nature of disagreement.

Purpose of using Discussion Method

Discussion as a method of teaching history may be used for the following purpose.

1. For preparing plans for new work
2. For making decisions concerning future action.
3. for sharing information and reaching a collective decision.
4. For tolerating and obtaining respect for various points of view
5. For clarifying ideas and concept
6. For inspiring interest
7. For evaluating progress
8. For stimulating thinking
9. To minimize the social distance between the teacher and the students.
10. To provide training for collective decision-making.

Procedure of the Discussion

Discussion may be informal or it may assume some such form as debates, a symposium or a panel discussion. Any one of these forms can be profitably adopted.

The procedure of Discussion may be divided into the following three steps.

1. Preparation for Discussion
2. Conduct of Discussion
3. Evaluation of Discussion

1. Preparation for Discussion

To conduct the discussion properly and effectively, through preparation for the same in a pre-requisite. The topic of discussion may be decided upon and the students may be provided with the names of the reference books and other reading- material from which the students may read the relevant material in order to take part effectively in the discussion.

2. Conduct of Discussion

This implies holding the discussion in a successful manner. The discussion should be systematic and disciplined. The arrangement of seats should be such as to permit face-to-face conversation. The teacher should see that everybody cooperates and takes part in the discussion. A few students should not be allowed to dominate the discussion. The discussion must result in some sort of an agreement.

3. Evaluation of Discussion

After the discussion, the main arguments raised during the discussion must be summed up. The advantages and the limitations must be thoroughly

weighted and evaluated. The outcomes of the discussion must be properly assessed, and lessons and guidelines must be drawn to make the discussion more effective and successful in future.

Role of Teacher

The success of the discussion method depends greatly on the teacher who is supposed to be a guide - cum - counselor - advisor. He should ensure that every student participates actively in the discussion, and makes his own contribution for the solution of the problem under discussion. He is also to maintain proper discipline, and see that the students should not go away from the topic under discussion.

The teacher is the leader of the discussion and must guide the students without using his authority. He should see that the discussion is a cooperative experience, not a competitive quarrel. He must discourage any personal attacks and must seek to bring the participants to focus their comments on the proposition and not on the person. The teacher is responsible for retaining control of the class and is responsible for retaining control of the class and is responsible for seeing that the discussion makes progress. Thus the success of Discussion Method mostly depends upon the teacher.

Advantages of the Discussion method

1. Enhances student learning.
2. Helps to motivate students.

3. Helps teachers to understand and evaluate students' thinking.
4. Helps to exchange ideas between teacher and students.
5. The consultation method is used to get the student to take the ideas without fear.

Disadvantages of the Discussion method

1. If the number of students in the class is 20 or more, this method will not be effective enough.
2. The class environment of the discussion method only a few students will dominate the discussion.

4.6 Socialized Recitation Method

According to Yoga and Simpson children discuss, question, report and do their work in a socialized way of comparison or discussion in a natural way. This is a teaching technique. In which friendly cooperation between students and teacher prevails. Children discuss various social, economic, cultural, political and moral issues and discover for themselves their true meaning and significance.

Debate is one of the most valuable techniques in teaching history. Aims to solve a problem by establishing consensus. There will be conflicting opinions and interests in the discussions. The two are a joint decision-making process. The discussion method may be extraneous or may have some form, such as a symposium or group discussion.

The process of discussion

The process of discussion can be divided into the following three steps.

1. Preparation for discussion
2. Conducting discussion
3. Evaluation of the discussion

1. Preparation for discussion

As a precondition for the complete preparation for conducting the discussion in a systematic and effective manner, the topic of the discussion may be determined and the books may provide re-learning materials for students to participate effectively in the discussion.

2. Conducting discussion

It means to conduct the discussion in a successful manner. The discussion should be systematic and ethical. The seat should be adequately folded. The teacher should note that everyone cooperates and participates in the discussion. Some students should not be allowed to dominate the discussion. The discussion should lead to some kind of agreement.

3. Evaluation of the discussion

After the discussion, the main points raised during the discussion should be summarized. Advantages and limitations should be fully weighed and evaluated. Guidelines should also be drawn up to make the discussion more effective and successful in the future.

The role of the author

Success in the discussion mode depends on the mentor and the teacher who is to be the counselor and counselor. The author is a mentor, consultant and director. Make sure every student actively participates in the discussion. The author is the leader of the discussion. The teacher should guide the students in the best way without using his power. Competition is not a fight. The teacher should not encourage any personal attacks. It is the responsibility of the teacher to ensure that the discussion is well conducted. So the success of the discussion depends on the author.

Benefits of Discussion

1. Helps students to overcome their fears and share their ideas with others.
2. It sets the stage for new thinking.
3. Course ideas help to think and act logically on social or economic issues.
4. Helps to share information and reach a collective conclusion.
5. Helps to make decisions related to future activities.
6. Excessive information can be obtained through discussions on a particular topic.
7. Students with the same interest become friends through discussion.
8. Helps to think well and make decisions before participating.

Benefits of Discussion

1. The discussion method is not suitable for all topics
2. It cannot be said that all students will participate in the discussions.

4.7 Source Method

The source method is one of the modern history teaching methods. There are many types of historical sources. Source method is the method of teaching using basic resources to teach History from material source, written source, oral source, pictures, photographs, caricatures, posters, history maps, maps, statistical results, sound and light sources, web site. The classroom is not a boundary for teaching modern history. Direct experiences gained outside the classroom also support teaching. The resource teaching method allows for field research, staring at historical monuments, visiting archives and museums with the author and getting to know the historical events material directly. In this way learning is active and the ability to focus prevails. The nature of the source of history is found here. The scientific method is the perfect method for History. But not for history. The historian would not have seen the events firsthand. So he uses his imagination to reconstruct with understanding. For this he looks at various sources and reads and studies.

Teaching and interpreting history using resources is considered to be similar to the laboratory method. Science does not invent anything new in the laboratory.

Instead students gain experiences about it while redoing what they have already discovered. Similarly students gain experience and skills when the teacher teaches the history lesson using resources. Those events come to life when taught using resources for the events in the textbook. Students know its authenticity when they look at the evidence. To make the lesson ideas real, visit the teacher's memorials, show the students the coins, ammunition, costumes, contracts, etc. in the museum and make it clear that the history lesson is related to real events.

History should be taught through sources such as archeological sites, museums and archives. Scheduling using different sources, comparing them with students. The teacher should make students capable of critically examining them and making appropriate decisions.

The importance of the source system

- The source method is used to analyze historical knowledge. The teacher or course book here does not give knowledge. Rather the learner learns knowledge on his own. So this is a learner-centered approach.
- Knowledge here is gained through various sources such as content, context and undertaking. The learner learns to use the actual source. Sees and learns to use available to us immediately. It is not available to us immediately upon seeing. Rather

than critique and getting the source, external reality grows.

- Research mindset prevails among students when analyzing originality.
- Learning becomes permanent, meaningful and robust as it is the first learning experience for the learner.
- In this way the learner becomes critically minded. Such skills can be used to solve problems later in life.

4.8 Unit Method

The unit method of instruction has evolved out of Project method, socialized recitation, lecture method etc. In fact that unit method includes about any technique or device that may be employed in helping pupils to grow. Its limitations in application and usefulness are not inherent in the method itself, but in the vision, enthusiasm, resourcefulness of those working with it.

We use the term unit to indicate any convenient division of subject matter, such as a topic, a chronological; division of subject matter, such as a topic, a chronological division or even a chapter. Such as division is merely convenience in organising the work, based on the amount that can properly be expected of pupils in a given assignment. But Jones, Grizzell and Grinsted issue a warning against any such concept and define a unit as

"A comprehensive and significant aspect of the environment, of an organised science, or an art or of conduct which being learned results in adaptation in personality".

Unit method for student's development

The unit method of instruction gives opportunity for pupils to work together in committee, to follow problems of their own interest and ability, to be creative to use their own initiative and to plan their own work under guidance. This method stresses learning, rather than teaching; pupil activity, rather than teacher activity. A good subject matter unit should include graphs charts, time-lines, map exercises, problems, projects, cartoons, posters, booklets, panel discussions, debates, skits, oral and written expressions, field trips, excursions, techniques of self-evaluation. In a well planned unit, dates, men, events and otherwise meaningless or detailed, materials all fit into a pattern where they have meaning and are learned in connection with a big movement or significant aspect of the environment. A unit should be organized around pupil activities, pupils experiences and opportunities for pupil development.

Morrison's unit plan includes five steps

1. **Exploration** : An action of investigation an unfamiliar or unknown area
2. **Presentation** : New idea or work is shown and explained to students
3. **Assimilation** : Process of fully understanding information and ideas.

4. **Organisation** : Explains the role of systematic planning in solving problems
5. **Recitation** : The act of reading or repeating aloud oral reply to questions.

The definite number of steps may be adopted to individual needs and interests. Some may be preparing a skit, others may clinch the idea in a cartoon, a poem, a story; others may prepare tables. Panel discussions or bulletin board displays. Help may come from text books, library books, other individuals and in some cases from studying community.

Uses of the unit system

1. This method encourages problem solving.
2. Promotes social and intellectual experience.
3. Develops oral presentation skills.

Disadvantages of the unit system

1. Students have to work hard.
2. This is a more time consuming method.
3. This places a heavy burden on the teacher.

4.9 Team Teaching

T - Together

E - Everyone

A - Achieves

M - More

Definition :

All arrangements that include two or more faculty in some level of collaboration in the planning and delivery of a course materials to a class. Team teaching has now found in different departments ranging from school to college level. It can help create a dynamic and interactive learning environment. In ordinary class teaching only one faculty covers entire course material which is interested or uninterested to the students.

But in team teaching course material is divided according to their specialisation and interest. Students have the opportunity to hear multiple perspectives. Different faculties create interest, keep attention and prevent boredom.

Types of Team Teaching

1. One teach - one observe
2. One teach - one Assist
3. Rotational Teaching
4. Parallel Teaching
5. Inter active Team Teaching

Let us see the above models in detail.

1. One Teach - One observe

This is the least collaborative approach to team teaching. This model places the responsibility for instruction on one individual usually the content area instructor while the other teacher floats around the

room making observations. These observations could be general or specific to a student.

2. One Teach - One Assist

This model is similar to the one teach - one observe model that a single teacher usually the content expert provides instruction to the whole class. The second teacher works directly with individual students. They may address either behaviour or academic questions or concerns.

3. Rotational Teaching / Station Teaching

Faculty members take classes in rotation. The class is divided into smaller groups and then these groups rotate through lessons by each instructor. Depending on the number of students in the class, more groups may be required to retain the advantage of small group instruction that this model offers.

4. Parallel Teaching

The parallel teaching model also has the benefit of smaller groups. In this model the students are split in half and each teacher teaches the lesson to one half the students. There is no difference in the materials presented to students by each teacher in this model. Each half of the class receives the same information and there is no rotation of students or teachers.

5. Interactive Team Teaching

Here two faculty members present in front of the class simultaneously and teaches a lesson one by one continuously. Continuation of the subject matter will be

presented by these two teachers till the end of the period.

Benefits for faculty from Team teaching

1. Improve their own teaching skills
2. Avoid the lonely repetitive fragmented experience of solo teaching.
3. Have opportunities for creative assignments.
4. Develop clearer perspective on the difference between disciplines.
5. Build bridges of understanding across disciplines.

Benefits for students from team teaching

1. Make classes more interesting and challenging because of the novelty.
2. Improve student learning outcomes, interpersonal skills, communication skills, analysis and judgement.
3. Team teaching allows students to involve high level intellectual debate among faculties.
4. Students have the opportunity to see different teaching skills and learning techniques.
5. The presence of another teacher in the class reduces disciplinary problems.

Disadvantages

1. Some teachers are rigid personality type or may carry to a single method of teaching.
2. Some simply dislike the other teachers on the team.
3. Allocation of course material to different faculties may sometimes be inconvenient.
4. Since two or more faculties are present in the class, expenses are more.
5. Teams don't have adequate common planning time.

4.10 Supervised Study

Supervised study sessions are conducted in schools before or after school hours. Most students from poor families come to school without adequate facilities at home to study. Some students are unable to study at home due to home environment, visit of relatives or TV shows. Therefore the supervised study session is used for such students to study the subjects without any hindrance and to clear up any doubts that may arise in them.

The teacher acts as a guide for the supervisory study. This supervised study takes place in the classrooms of the school.

Classifications of supervisory study

1. Two course program

Teach one lesson here and the next lesson is used for supervised study in the lessons taught by the teacher.

2. Plan to divide the course

The course can be divided into two halves and the first half can be used for teaching and the next half for supervised study.

3. Plan for more daily lessons

This is to set aside an extra course at the end of each day and conduct a supervised study in it.

4. Library study program

Here one lesson per week can be set aside for the library and during that course can be set aside for reading library books. The librarian or course teacher will supervise the study.

Techniques of supervised study

1. The success of the supervisory study depends on the teacher who teaches the students how to study.
2. Students' study room should provide a readable environment.
3. Care should be taken not to distract while reading.
4. Students should bring books and notes to read.
5. The teacher should explain the method of engaging in low-level reading (Skim Reading) to know the general concept of a subject.

Benefits of the supervisory study

1. Be able to get immediate help and advice during the supervision course.
2. Able to read school lessons properly.
3. Used to spend leisure time effectively doing homework and self-study.
4. In this way the relationship between teacher and student is strengthened.
5. Helps to assess students' ongoing assessment and improvement.

Disadvantages of supervisory study

1. The teacher is not able to focus on the students because of the large number of students.
2. The workload of teachers increases.
3. Some students are unable to attend the supervisory course due to various reasons. This causes some problems in maintaining order.
4. The supervisor sometimes gives punishments to students who make mistakes. Thus undermining the purpose of the supervisory course.
5. Students get tired when they engage in learning activities for an additional hour after school.

4.11 Programmed Instruction

Programmed Instruction is also known as programmed learning as learning is focused in it. It provides opportunity to the learner to study and learn individually.

Learning opportunity is provided through a frame or series of frame having i) learning material ii) in built mechanism of evaluation and iii) related answer of the evaluation part to provide feedback. It is considered the antecedent of modern computer-aided learning.

Principles of programmed instruction

- Principle of small steps
- Principle of Active responding
- Principle of immediate feedback
- Principle of Self-Pacing
- Principle of Student Testing

Programmed Learning

1. **Principles of small Steps** : This principle is based on the basic assumption that a person learns better if the content matter is presented to him in suitable small steps and in meaningful segment of information called frames.
2. **Principles of active responding** : The principle rests on the assumption that a learner learns better by being active. In programmed learning, the learner may remain active if he responds actively to every frame presented to him. Therefore a good

programme should actively involve the learner in learning process.

3. **Principle of immediate reinforcement** : One person learns better when he is motivated to learn by receiving information of the result just immediately after responding. Therefore in a good programme, appropriate consideration is always made for the provision of immediate reinforcement by informing about the correctness of his response.
4. **Principle of Self-pacing** : Programmed instruction is based on the basic assumption that learning can take place better if an individual is allowed to learn at his own pace. The programming of the material should be done in view of the principle of individual difference and the learner should be able to respond and move from one frame to another according to his own speed of learning.
5. **Principle of student-testing**: For better learning, it is important to seek continuous evolution of the learning process. The principle of student testing meets this requirement. In the programmed learning, the learners record of his response and its feedback is immediately given.

Example of a frame

In the multiplication of $3 \times 4 = 12$, the number 12 is called a _____

- a) Factor b) Quotient c) Product d)
Power

In the answer is _____

- a. Factor (Branch to remedial X_1)
- b. Quotient (Branch to remedial X_2)
- c. Product (Reinforce, go to next frame)
- d. Power (Branch to remedial X_3)

Programmed instruction is a method of individualized instruction where each individual learns by himself at his own rate. Programme is a sequence of small steps of instructional material (called frames) most of which requires a response to be made by completing a blank space in a sentence. A system of cueing is applied and each response is verified by the provision of immediate knowledge of results. The teacher is not physically present. The programmer while developing programmed materials has to follow the laws of behavior and validate his strategy in terms of student learning. It is based - on skinner's operant conditioning.

Characteristics of Programmed instruction

- It is an individualized instruction
- Materials are developed keeping in view logical sequence
- Immediate feedback is being provided
- There is inbuilt mechanism of interaction between the learning materials and learners.
- Learners learn the material with their own pace
- Continuous evaluation is integral part of the learning material.

Development of Programmed Learning Material

- i. Preparatory phase
- ii. Development or writing phase
- iii. Evaluation phase / Tryout and Revision

i. Preparatory phase

- Selection of a unit or topic
- Preparation of content outline
- Defining objectives in Behavioural terms
- Construction of a Test of Entering behavior
- Construction of Test of Terminal behavior

ii. Development or writing phase

- Present the material in Frames
- Provide for Active Student Response
- Constructing test after each frame
- Provide for correction of student Response
- use of prompts to Guide student Response
- Provide careful sequencing of Frames
- Preparing answer Key

iii. Evaluation phase

- To evaluate to improve through pre-tryout and tryout.
- To Revise
- To evaluate the impact of the learning material

Advantages of programmed Learning

1. There is no fixed time for learning.
2. Students may learn at their own pace.
3. Programmed learning is an strategy for self-learning easily without any tutor.
4. It gives drill, practice, self-evaluation and motivation
5. It is a new method of presenting context to students through consequent sequences in a controlled manner.
6. It enhances students critical thinking ability and power of judgement.
7. In short period of time students can learn huge content.
8. It helps for individual learning and avoids social and emotional problems.
9. It immediately provides result of progress
10. It is very helpful for distance learning.

Disadvantages of programmed Learning

1. It limits students in creativity and originality
2. Preparation of material is time consuming
3. It restricts student teacher interaction and relationship
4. Students may get separated from society
5. Doubts cannot be resolved through this instruction immediately as formal education.

4.12. Computer Assisted Instruction (CAI)

Computer Assisted Instruction is an interactive instructional technique whereby a computer is used to present the instructional material and monitor the learning that takes place. It may be used in offline / online. It is a self learning technique whereby the computer is used to provide learning resources, record keeping, progress tracking and assessment of learner performance.

CAI uses a combination of text, graphics, sound and video in enhancing the learning process. The computer has many purpose in the classroom, and it can be utilized to help a student in all areas of the curriculum. CAI helps to improve instruction. CAI programs are tutorials, drill and practice, simulation and problem solving approaches to present topics and they test the students' understanding. They will also give notes on incorrect responses. Generally it involves three types of techniques namely hardware, software and courseware

Types of Computer Assisted Instruction

1. **Drill and practice** : Drill and practice provide opportunities in skills to become mastery in the content.
2. **Tutorial** : Tutorial activity includes both the presentation of information and its extension into different forms of work, including drill and practice games and simulation.

3. **Games** : Content in the forms of Games are presented Games software help to achieve highest score.
4. **Simulation** : Simulation software can provide an approximation of reality.
5. **Discovery** : Discovery approach provides a large database of information specific to a course or content area and challenges the learners to analyze, compare, infer and evaluation based on their exploration of the data.
6. **Problem Solving** : This approach helps students develop specific problem solving skills and strategies.

Advantages of CAI

- CAI provide one-to-one interaction
- It is a great motivator
- Instaneous response / immediate feedback to the answer elicited.
- Self-pacing - Allow students to proceed at their own pace.
- Helps teacher can devote more time to individual students
- Privacy helps the shy and slow learner to learn
- Learn more and more rapidly
- Multimedia helps to understand difficult concepts through multi sensory approach

- Self directed learning - Students can decide when, where and what to learn.
- CAI can be arranged for nearly 4000 students simultaneously.

Disadvantages of CAI

- May feel overwhelmed by the information and resources available.
- Over use of multimedia may divert the attention from the content.
- Learning becomes too mechanical
- Non availability of good CAI packages
- Lack of infrastructure
- CAI can become boring if a student is alone at a terminal for too long time

4.13 The Keller Plan

The Keller plan, also called the Personalized System of Instruction (PSI) was developed by Fred. S. Keller and his associates in the middle of 1960 as an innovative method of Instruction. This individualized learning method was oriented on improvement of high school learning. It is highly personalised learning method in which students progress at their own pace. It is based on Skinner's theories of operant conditioning.

- Go-at-your-own-pace. A student can move through the course content at his own pace, independently of other students.

- Detailed definitions of learning objectives.
- Unit perfection requirement. In order to advance to the next unit, a student-needs to demonstrate mastery of the preceding unit.
- Emphasis on written materials. The emphasis is on learning from written materials. Lectures and demonstrations will be provided only when you have interest in it. No examination will be based upon them; and you need not attend them if you do not wish.
- The use of proctor. Proctors entrance the social aspect of education process, enable repeated testing with immediate scoring and tutoring. They help with individual problems.

Course module

Physically the teaching Lab has tables with seating for about 20. The lab is open to students for three to four hours each day and one or more tutors are available in the lab each of those hours. Students come to the lab to write their quiz and to seek help with course material from the Tutors or from other students. In the Lab students receive quiz sheets and work individually to answer them. Students may consult their texts, notes and calculators freely during the quizzes, monitoring of the students is not a concern. The quizzes consist of ten multiple choice questions and students must answer eight correctly. Otherwise they must return to the lab for the next attempt. If necessary students take repeated attempts to demonstrate their mastery. Multiple forms of quizzes are prepared for

repeated candidates. Result of the quizzes are posted in the lab on the next day.

The modules can consist of reading assignments, films, audio tape, film strips, programmed instruction, conducting an experiment, conducting an interview etc.

Keller divided the process for creating PSI into four steps.

- Determine the material for the course
- Divide the material into self-contained module (segments)
- Create methods of evaluating the material in a given module.
- Allow learners to move from module to module at their own place.

Advantages of Keller Plan

- PSI has significantly positive effects on learning.
- In Keller plan there is no punishments, and the instructors simply award pass or fail grades. It yielded better results than more traditional lecture-based learning.

Disadvantage of Keller Plan

- Limited instructional methods
- High dropout rates
- Decreased human interaction.

4.14 Project Method

Project is a purposeful activity related to life and it should be carried out in a natural environment in a school or outside the school. Project method is of American origin and is an outcome of Dewey's philosophy of pragmatism. This method is developed and applied practically by Dr. Kilpatrick. In this method student's learn subject matter in natural setting. Here the content knowledge and skills found in mathematical text book are carried out with the actual happenings in life. Projects can be done individually or in a group or all the students together .

Basic principles of project method

- Learning by doing.
- Learning by living.
- Learning through association, co-operation and activity.
- Learning by experience.
- Learning by sociability

Some projects for mathematics

- Running a co-operative bank in the school.
- Running a stationary store in the school.
- Running a school hostel.
- Lying road in the school.
- Maintaining kitchen garden in the school.

Steps in a project

For completing a project, we have six stages

1. Providing a situation.
2. Choosing a project.
3. Planning of the project.
4. Executing the project.
5. Evaluating the project.
6. Recording all activities of the project.

1) Providing a situation:

A project should arise out of a need felt by students and content of the subject. It should never be forced to them. For this teacher will explain various projects according to need, interest of the students, and their skills to do the project.

2) Choosing a project:

Project will be selected according to the wish of the students. The teacher may give guidance in selection of a better project if the student's make an unwise choice.

3) Planning the project:

The students he encouraged by the teacher to plan out the details of the project. While planning, the following points to be considered. i) the nature and scope of the project. ii) the degree of complexity of project. iii) time allotted to finish the project and iv)

availability of material resources. Discussion may be held among students before the final decision is agreed upon.

4) Executing the project:

The teacher helps the student's in assigning work to different student's accordance with their interest, aptitude and capabilities. Each member of the group should be actively involved in the execution of the project. The teacher supervise and guide the students in the execution of the project.

5) Evaluation

The students along with the teacher should review the progress of the project at frequent intervals. The evaluation of the project has to be done in the light of plans, difficulties in the execution and achieved results. Let the students have self criticism and look through their own failings and findings.

6) Record

A complete record of the project be kept by the students. The record should include everything from the beginning to end about the project. It should include the proposed plan, duties allotted to different students and how far were they carried out by them, details of places visited, guidance we received, books consulted and all other details.

Example 1: Laying out a vegetable garden in the school.

Providing a situation: After knowing the interest, abilities of the students and content. Teacher explain projects like laying road in the school, Lying vegetable garden, Running a stationery store.

Choosing a project: According to wish of the students the project selected is laying out a vegetable garden in the school.

Planning the project: To lay a vegetable garden in the school, the following points are discussed.

- i. Area of the garden and place selected for garden.
- ii. What are the vegetables to be grown ?
- iii. How much of seeds, fertilizers and pesticides we have to buy?
- iv. Type of work to be allotted to students.
- v. Total amount required to form the garden.
- vi. Expected revenue.

Execution: Each member of the group actively involved in the execution of the project. The teacher superwise the work and guide the students then and there.

Evaluation: With the help of project, students learn how to calculate the area of the garden, profit or losses and percentage of profit or loss.

Recording: All activities of the project are recorded here. Area of garden, area of vegetables planted, total expenditure, total revenue, profit or loss, profit or loss percentage guidance for future etc are all recorded here.

Example 2:

Running of a hostel mess in the school

It involve the following steps.

1. The number of hostlers will be recorded.
2. Secretary of the hostel will be selected by the hostlers.
3. The expected expenditure will be calculated.
4. Expenditure on various heads will be allocated.
5. Budget will be prepared.
6. The amount of collections from the students will be noted.
7. A chart of 'balanced diet' for the hostlers will be prepared.
8. The time of breakfast, lunch, tea and dinner will be notified.
9. Weight of each hostler will be checked after regular intervals and the same will be put on notice board.
10. Punctually in all the activities will be maintained.
11. Evaluation of all the entire program.
12. Entire activities are recorded and will be informed to all concerned.

Merits of project method

1. It is a psychological method and learner's centered.
2. It introduces democracy in education.

3. It promotes social interaction, inculcates spirit of co-operation and exchanges of experiences among the students.
4. It develops self confidence and self-discipline.
5. It makes the learning more interesting and facilitates better understanding of the subject matter as the learning is related to reality.

Demerits

1. The project method is expensive and time consuming and it is not possible to fit into the regular timetable.
2. It does not provide any training in mathematical thinking and reasoning.
3. Systematic and continuous teaching is not possible.
4. The knowledge obtained from various activities are not continuous.
5. All the students are not getting same type of learning.

4.15 Activity Based Learning (ABL Method)

To make students to involve in an activity and thereby learning the subject matter with interest is called Activity based learning. Children with the help of materials participate in activities, doing exercises given and fulfill in learning.

In Activity based learning a skill is taken and learn it without any doubt and then only be proceeds to next skill. After knowing all skills of a class, he will be

allowed to go next class. Children themselves take the learning cards of a skill and learns step by step.

For the first time in Tamilnadu activity based learning was introduced in 2003 in 13 model schools under 10 regions of Chennai Corporation. Due to success of this method, this method was introduced in all schools of Chennai corporation in 2004-05. From the academic year 2006-07 this method was implemented all over Tamilnadu in elementary schools for the stds 1 to 4. The implementation of this method in Tamil nadu is due to credit of Thiru M.B Vijaykumar I.A.S.

Approaches of Activity Based Learning

- Following the curriculum completely.
- Each skill is divided into small parts and change it into learning steps.
- In each learning step various activities are followed.
- All learning steps are linked to complete the syllabus.

Activities arranged in each learning step

- Initial activities are formed for skill development.
- Giving motivation for the skill introduced.
- Activities for skill exercises.
- Evaluation activities.
- Diagnostic activities.
- Wholesome activities for the skill.

The four core learning styles of Activity based learning are

1. Visual
2. Auditory
3. Reading and writing
4. Kinesthetic

1. Visual

Visual learners are better able to retain information when it is presented to them, such as arrows, charts, diagrams, symbols and more.

2. Auditory

Auditory learners prefer listening to information that is presented to them vocally. These learners work well in group settings where vocal collaboration is present and may enjoy reading aloud to themselves too.

3. Reading and Writing

Focusing on the written word, reading and writing learners succeed with written information on work sheets, presentations and other text-heavy resources. These learners are note takers and perform strongly when they can reference written text.

4. Kinesthetic

Taking a physically active role, kinesthetic learners are hands-on and thrive when engaging all of their senses during course work. These learners tend to work well in scientific studies due to the hands-on lab component of the course.

Following are few activity based learning methods

1. Real objects :

Real objects work better when trying to understand them rather than something virtual or imaginary. Science lessons can be extremely interesting through this method of teaching. For example let us go for classification of plants. Students can be asked to collect different varieties of plants and the teacher can teach them about the plant during class.

2. Project work :

For projects, students do it themselves or get help from their parents. But whatever it may be, students get to familiarise with the project topic they are working on. Having regular project work not only increases subject knowledge but also increases curiosity to learn more and with clarity.

3. Video classes :

Videos have taken the centre stage in grabbing attention. Any form of video will earn the interest of the students. Even a short video featuring news could be the subject of interest for students who are bored of seeing the blackboard all the time. Arrange video classes to teach the content.

4. Field trip :

Many a times sitting in the same classroom is boring. Changing the environment can do wonders. If you want to teach post-office, railway station, river or a factory take the students to that place and teach them.

5. Story and role plays :

Do you now why language classes are more interesting than the other subjects? It is because of lots of stories and enacting them through role plays. This method could be tried out for other subjects too. It helps in breaking the monotony of the lessons and getting the students animated.

Merits of Activity Based Learning

1. There is no content burden.
2. No examination fear for children.
3. Learns on par with individual difference.
4. Teacher - student relation improves.
5. Learns with peer group.
6. Students who are absent can continue their lessons where they left.
7. Improves with self - Confidence.
8. Backward students improves in learning with the help of gifted students.
9. Teacher can teach speak with individual student.
10. Flexible home assignments.
11. Recognition is given to particular achievements.
This create interest to students to learn further.
12. This is a curriculum which develops full personality.

Demerits

1. Teachers work load increases.
2. If the number of students in the class is more than class discipline fails.

3. Learning affected by a teacher who took frequent leave.
4. Evaluation of learning is difficult.
5. Since we are teaching only minimum skills which are required, we are not competent with private schools.

4.16 Active Learning Method (ALM)

Active learning is an approach to instruction that involves activity engaged students with the course material through discussions, problem solving, case studies, role plays and other methods. Active learning approaches place a greater degree of responsibility on the learner than passive approaches such as lectures, but instructor guidance is also necessary in the active learning method. Active learning activities may range in length from a couple of minutes to whole class sessions or may take place over multiple class sessions.

Active learning activities help promote higher order thinking skills such as application of knowledge, analysis and synthesis. These activities engage students in deep rather than surface learning and enable students to apply rather than surface learning and enable students to apply and transfer knowledge better.

Principles of Active learning

1. Reflective - Students' reflection on the meaning of what is learned.
2. Negotiated - negotiation of goals and methods of learning between students and teachers.

3. Critical - Student appreciate different ways and means of learning the content.
4. Complex - Students compare learning tasks with complexities existing in real life and making reflective analysis.
5. Engaged - real life tasks are reflected in the activities conducted for learning.

Nature of active learning

Active learning techniques include role-playing, case studies, group projects, think-pair-share, peer teaching debates, teaching and short demonstrations followed by class discussion. In active learning the teacher acts as a guide. The teacher's task is to let the learners learn on their own through the use of different active learning activities. Active learning is the opposite of passive learning. It is learner - centred not teacher - centred. Active participation of each and every student is a necessary aspect in active learning.

Active Learning Techniques

Few active learning techniques are given below :

1. Tyre Think-Pair-Share activity encourage all students to interact with the material. In this activity, the instructor states an open-ended question. Ask students to spend a minute or two thinking about the writing a response. Then ask students to pair with a partner to discuss their responses.

2. Use a one minute paper test in your class as a formative assessment. Give students 1 to 2 minutes to write brief responses. Address student responses either during next class or online.
3. Providing Feedback : Students often don't know why something is wrong, so whenever possible and time permitting, take a few moments to explain why you have marked incorrect on tests and assignment. Give plenty of positive feedback as well as feedback
4. Cooperative learning : Students learn effectively when they are working together. Plan activities that require students to work together and learn from one another. In the process they will learn critical thinking skills, communication skills, problem solving skills and more.
5. Class Discussion : Another way for students to teach each other is through class discussion. As students take turns discussing the subject, you can assess their knowledge and discover which students grasp the concept and to what extent.

Advantages of active learning methods

1. Research has proven that there is increased content knowledge for participants of the approach.
2. Development of critical thinking and problem-solving are other benefits.
3. Creative thinking, collaborative and interpersonal skills also show great improvement when active learning methods are implemented.

4. Students engage in higher-order thinking tasks such as analysis, synthesis and evaluation.
5. Promote research based learning through investigation
6. Encouraging leadership skills of the students through self development activities.
7. Integration of prior knowledge with new knowledge to incur a rich structure of knowledge among the students.
8. Cultivating a dynamic environment through inter disciplinary learning and generating high- profile activities for a better learning experience.

Activities - based Learning principles for the Digital Classroom

Due to Covid-19 pandemic schools, colleges and universities across the country were forced to accelerate their on live lesson delivery. There are five activity - based learning principles, teachers can bring into their digital delivery.

1. Act as a guide rather than a lecturer

No one wants to state at a talking head for hours. Break class time into short segments. Between each segment provide students with activities such as polling questions or clicker questions that students can answer with their own smart phone. Other ideas for between segment breaks are lecturer provide as on topic quote and students take five minutes to formulate an argument for or against the topic.

2. Maintain open Communication with students

Since lecturers are generally teaching to a webcam, they have no way of picking up the visual clues from students. Using calls and emails, get how well students are learning. suggested activities include, inviting students to design and lead class discussion, asking students to email the instructor a question or idea related to the topic.

3. Encourage student interactions

When students learn only on online, they miss out on learning form their peers. You can implement some strategies to facilitate interaction and get students to exercise their communication skills during and when outside of class session.

4. Be flexible and creative

With the shift to online learning, boredom and tensions can run high. Try to inject some humor into lessons. Be as flexible as you can with your assignment time table.

5. Explore interactive online tools

There are variety of online tools for interactive teaching. Zoom, Panopto, Google hangout, skype and pre-recording teaching resources etc are interactive online tools use the available resources for the students.

4.17 Mind Map

A mind map is a diagram in which information is represented visually, with a central idea placed in the middle and associated ideas arranged around it.

Mind mapping is a way of linking key concepts using images, lines and links. A central concept is linked via lines to other concepts which in turn are linked.

Buzen (1993), the inventor of mind maps, claimed that mind mapping is vastly superior to traditional note-taking methods.

A mind map is a tool for the brain that captures the thinking that goes on inside your head. Mind mapping helps you think, collect knowledge, remember and create ideas. Most likely it will make you a better thinker. Mind maps can be created in many different ways, but they share the same basics. Mind map softwares are also available.

Purpose of mind map

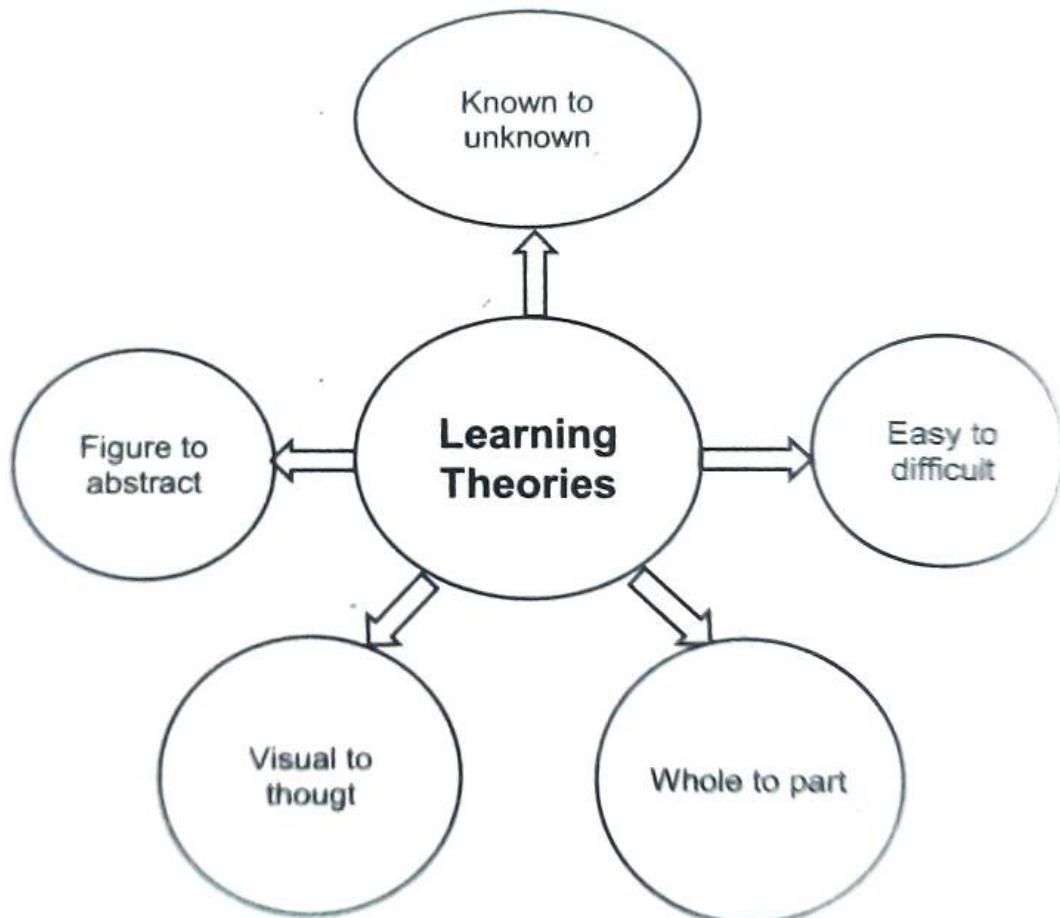
- To present information effectively
- To improve reading comprehension
- To inspire creativity
- To improve memory and recall

Steps to draw a mind map

1. Write the main idea in the center of the paper

2. Draw branches that point away from the center. Each branch symbolizes one thought or idea related to the idea of the subject.
3. From each branch more ideas can branch off.
4. If necessary use colors and images whenever possible. Research shows that using colors and images are better in remembering than those who do not.

Mind Map - Example



Uses of Mind Map for Teachers

- Mind mapping for lesson plan can help teachers or trainers identify teaching route and increase recall of

the subject matter. This can boost teaching confidence.

- As a Pedagogical tool, mind map provides an effective approach for promoting better understanding in students.
- It is a useful strategy for introducing new concepts and research projects.
- A teacher must design for class curriculum for the school year with the help of mind map and gives her a clear and overview of what needs to be cover.
- By using mind mapping to plan her teaching, a teacher can reduce the amount of notes she takes into class
- Mind maps are ideal for teaching and presenting concepts in the classroom as they provide a useful focus for students.
- Mind maps can be used in class to brain storm and generate discussions.
- Mind map are a great tool to present complex concepts.
- It is a useful tool to present information clearly and in logical order.

Uses of Mind map for Students

- This learning technique promotes greater creativity for all learners.

- Mind mapping was developed as an effective method for generating ideas by association.
- Mind mapping can be used for assignments and essay writing especially in the initial stages, where it is an ideal strategy for thinking.
- It can also be used for generating, visualizing, organizing, note-taking, problem-solving, decision-making, revising and clarifying a course content.
- Mind mapping raise the performance of students at all levels of ability as they become more efficient in generating and organizing ideas for their writing.
- The students also displayed a positive attitude towards using mind-mapping as a pre-writing activity.
- It is also for easier for student to remember single words in mind map rather than to remember long sentences.
- By using key words in a mind map, a student opens up his or her thinking and stimulate his or her mind to dig deeper and see greater detail on thoughts.
- Mind mapping enhances students' critical thinking and co-operation as well as problem solving.
- Mind mapping can be helpful for all kinds of educational tasks such as note-taking, creating, engaging presentations and more.

- Taking notes in a lecture and listening for the most important points.
- Showing links and relationships between the main ideas in the subject.
- Stimulating creative thinking and creative solutions to problems
- Reviewing learning in preparation for a test or examination.
- Mind maps are more compact than conventional notes.
- Mind Mapping can also help you to break down large projects or topics into manageable pieces.
- Mind maps are unique for individual students.

4.18 Advanced Active Learning Method (AALM)

Advanced Active Learning method is designed to engage a students' critical thinking skills and apply previous and new knowledge to real-life situations. It emphasis on the higher order learning processes.

It is designed for students for their own learning with structured and open-ending guidance from instructors.

Importance of Advanced Active Learning Method

Advanced active learning involves student development of critical thinking and problem solving skills. More advanced engaged learning activities that

require additional planning by the instructor are case-based learning, team - based learning or problem-based learning. Students continue to fail in their ability to demonstrate complex thinking skills because their educational experiences have provided little support for development of these skills for maximal performance. Multiple teaching styles exist to encourage students to apply critical thinking skills to open-ended problems utilizing multiple points of view.

Responsibility of the Instructor

The traditional model of students are passive recipients of learning which is to be inadequate to factor a students understanding and enhance a students' interest. As students' become active learners, faculty become active teacher. A teacher's responsibility moves beyond providing a educative lecture and more towards encouraging students to be more independent after providing a conducive learning environment.

Advanced Active Learning Strategies

Case based learning or scenario-based learning helps students develop a deeper understanding of the material. The instructor is responsible for creating a complex and engaging scenario or case which parallels the concepts of the lesson. Student may be additionally directed using guided questions. Students will use critical thinking skills to analyse the scenario while communicating a collaborating as a group. Using this

technique, students are presented with real life scenarios that are multifaced.

Team-based Learning (TBL)

This approach utilizes student-student interaction in small team to develop critical, practical and creative teaching in their courses. Students assume the role of bring inquiries and faculty burnout is decreased with increased students responsibility and student engagement. Key characteristics of TBL include the creation of heterogeneous work groups.

The team based learning module is especially appropriate in larger class sizes by allowing the class to function or multiple small classes.

Problem-based learning

In this approach, a student learning is facilitated by an instructor presented a problem. Students will work through this carefully constructed open-ended problem by engaging and applying new knowledge. The success of PBL is dependent on several factors: Problems should allow for free inquiry, collaboration is encouraged and essential and feed back in the form of reflection on their learning should be provided. The premise of PBL resides in the students to apply previous knowledge to new situations.

For the success of Advanced Active Learning Method

- Allow the student to be the inquirer.

- Think strategically of what your students to accomplish by the end of the class and design your advanced active learning lesson accordingly.
- Be mindful of the time for preparation for an advanced active learning structure. If designed poorly, students may be less engaged.
- Employ an advanced learning model strategically that best fits the structure and purpose of the course.
- Heterogeneous groups should be created with a structure in place to allow each team to function independently.

REVIEW QUESTIONS

1. Explain any two teacher centred methods of teaching.
2. Explain any two learner centred methods of instructions.
3. Explain in detail the Keller Plan.
4. Explain in detail the Mind Map.
5. Explain in detail the Advanced Active Learning Method.
6. What is Mind Map? How it is useful to teachers and students.